RETI NEURALI RICORRENTI

MLP $\longrightarrow$ INPUT $\underline{x} \in \mathbb{R}^N$, N finito $\quad x$

CNN $\longrightarrow$ INPUT $I \in \mathbb{R}^{H \times W \times c}$ dove H, W, c sono finiti $\quad I$

DATO SEQUENZIALE

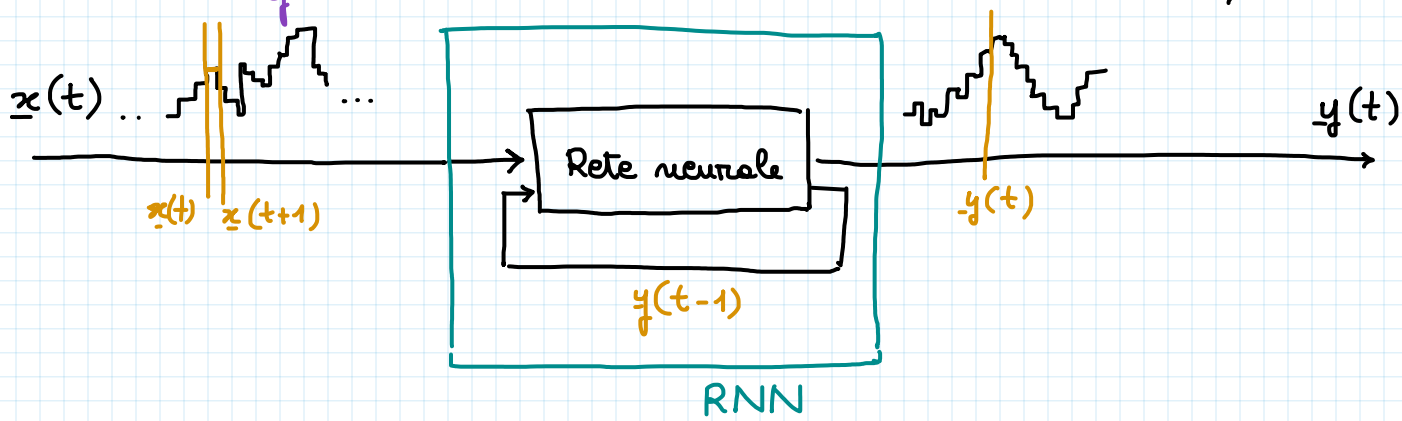
↳ INPUT : musica, video, serie temporali, testo

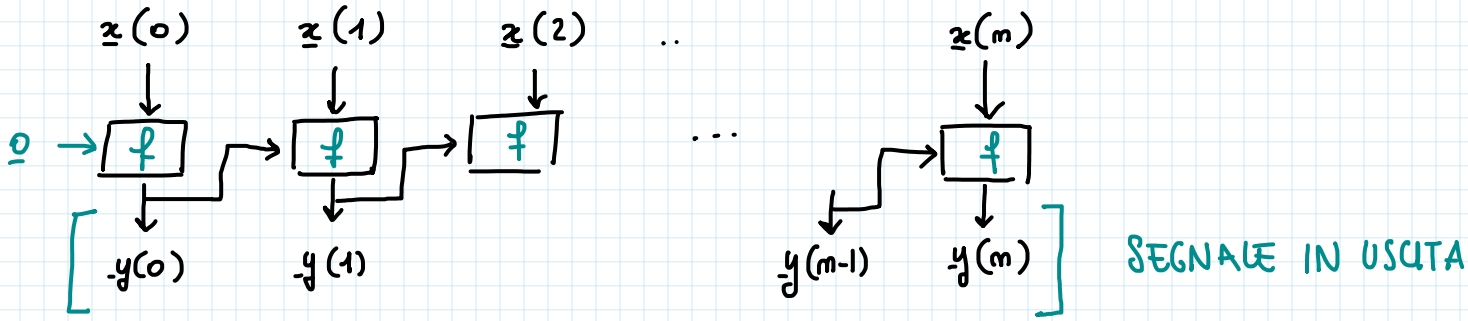
↳ OUTPUT : • speech synthesis IN/OUT

• image captioning OUT

• generazione di testo

PROBLEMA: Gestire la DIMENSIONE VARIABILE t ↪ CICLI





$$\underbrace{\sigma\left(W \cdot \underline{x}(t) + \hat{W}\, \underline{y}(t-1) + \underline{b}\right)}$$

$$\sigma\left(W^2 \cdot \boxed{} + \underline{b}^2\right) \quad \ldots$$

$$\sigma\left(W \cdot \underline{x}(t+1) + \hat{W}\, \underline{y}(t) + \underline{b}\right)$$

A t finito

$\left.\begin{matrix} \underline{x} \\ \underline{y} \end{matrix}\right\} \in \mathbb{R}^N$ dim fissa